Conservative Test-Time Adaptation under Open-World Contamination via Binary Correctness Feedback

Wenzhang Du

Dept. of Computer Engineering, Mahanakorn University of Technology International College (MUTIC), Bangkok, Thailand
dqswordman@gmail.com

Abstract. Streaming test-time adaptation can be viewed as an iterative online heuristic: each incoming window suggests a possible model update, and the deployed learner must decide whether that update should be accepted, rejected, or frozen. We study this update-control problem under open-world contamination. Our focus is not whether binary feedback universally improves mean accuracy, but whether queried paired binary correctness supervision on a small subset changes what can be controlled safely relative to purely unlabeled adaptation. We instantiate this idea as a binary-feedback-gated wrapper around entropy-minimizing adaptation. The gate evaluates a candidate update by comparing queried binary correctness before and after that update on the same queried subset, and accepts only when the empirical paired gain is sufficiently positive. A stylized analysis isolates the core boundary: unlabeled evidence alone can be ambiguous about update safety, whereas binary feedback yields a finite-sample acceptance test. Empirically, stationary CIFAR-10-C and CIFAR-100-C do not show a universal binary-feedback win. Under CIFAR-C plus SVHN contamination, however, the proposed gate preserves positive gain over freezing while removing the observed negative-transfer events that remain for Tent and matched 0-bit controls at the hardest settings in our main grid. The contribution is therefore a conservative binary-feedback veto primitive, not a general replacement for strong label-free test-time adaptation.

Keywords: test-time adaptation · contamination · binary feedback · online adaptation · conservative learning

1 Introduction

Test-time adaptation (TTA) updates a source-trained model after deployment in response to target-stream evidence. Strong label-free methods such as Tent, SAR, and COME show that entropy-style or stability-oriented updates can recover large gains on standard corruption benchmarks [12,10,16]. However, these benchmarks are largely judged by mean accuracy under stationary shift. For deployed online systems, that is not the only question. Under contamination, the central issue is whether the learner has enough information to decide whether a proposed update should be committed, rejected, or frozen.

We view streaming TTA as an iterative online heuristic. Each incoming window proposes a possible update to the deployed model. In contamination-free settings, aggressive adaptation can be beneficial. Under open-world contamination, the same internal confidence or entropy signal may correspond either to a genuinely helpful update or to a harmful one. This makes update control, not only adaptation strength, the central problem.

This paper studies that problem through an information-regime lens. In the label-free regime, the learner updates from target inputs and its own predictions only. In the binary-feedback regime, it may query a tiny subset and receive queried binary correctness supervision. Our claim is deliberately narrow:

1. purely unlabeled evidence does not in general determine whether an update is safe relative to freezing under contamination;
2. queried binary correctness supervision can support a conservative update veto;
3. this primitive expands the empirically safe operating region even when it does not maximize mean accuracy.

We instantiate this idea as a binary-feedback-gated wrapper around Tent. At each window, the method proposes an unlabeled entropy-minimizing update, evaluates queried binary correctness before and after that candidate on the same queried subset, and then accepts, rejects, or freezes. The point is not to replace strong 0-bit TTA with a richer adaptation objective. The point is to study whether small external feedback changes the update-control problem.

The empirical story is correspondingly bounded. On held-out stationary CIFAR-10-C and CIFAR-100-C, strong 0-bit baselines remain best on mean accuracy, so our results do not support a universal binary-feedback win. Under CIFAR-C plus SVHN contamination, however, the picture changes. At the hardest settings in our main grid, the proposed gate is not the highest-mean method, but it preserves positive gain over freezing while removing the observed negative-transfer events that remain for Tent and matched-conservatism 0-bit controls.

The paper makes three contributions. First, we state a clean ambiguity result showing why unlabeled evidence alone cannot generally certify update safety under contamination. Second, we connect the implemented paired binary gate to a finite-sample acceptance guarantee. Third, we provide a shared-protocol empirical study with matched controls, a second benchmark family, robustness analyses, and a supervision-accounting view derived entirely from existing evaluation runs.

2 Related Work

Label-free test-time adaptation. Unlabeled test-time adaptation aims to improve deployed models without access to target labels. Test-Time Training (TTT) converts each test example into a self-supervised update problem [11]. Tent later popularized entropy minimization with batch-normalization updates as a simple and effective label-free baseline on corruption benchmarks [12]. MEMO combines adaptation and augmentation-based marginal entropy minimization in local test-time settings [15]. Our work is adjacent to this literature but asks a different question. We do not propose a new label-free objective. Instead, we study when unlabeled evidence is insufficient for deciding whether a proposed online update should be committed under contamination.

Stable and conservative test-time adaptation. A second line of work tries to make label-free online adaptation more stable. EATA, SAR, CoTTA, and COME each modify the update rule or the update schedule to reduce collapse, forgetting, or error accumulation in changing environments [9,10,14,16]. These are the closest 0-bit comparators to our setting because they address the same practical concern: how to adapt without drifting too far from a safe source state. Our paper differs in the source of control. Rather than relying only on internal entropy, restoration, or confidence heuristics, we ask whether queried binary correctness supervision can serve as an external veto on candidate updates.

Feedback-assisted and active test-time adaptation. Recent work also studies settings with queried supervision at test time. ATTA analyzes active test-time adaptation with queried labels [3]. Effortless active labeling studies low-label long-term adaptation [13]. BiTTA is the closest existing feedback-based reference because it uses binary feedback rather than full labels [7]. Our goal is narrower than these low-label adaptation papers. We do not aim to show that binary feedback dominates richer supervision on mean accuracy. We instead study whether binary correctness feedback is already sufficient to support a conservative accept/reject/freeze primitive under contamination.

Open-world and contaminated test streams. Common corruption benchmarks such as CIFAR-C focus on stationary label-preserving shifts [5]. Continual and dynamic target domains have motivated methods designed for non-stationary TTA [14,10]. More broadly, open-set recognition work emphasizes that out-of-class or unknown inputs can invalidate closed-set confidence heuristics [1]. Our contamination setting is related to, but distinct from, open-set recognition. We do not study unknown-class rejection as the primary goal. We study how out-of-label-space contamination changes the safety of online test-time updates.

Online decision under limited feedback. At a higher level, the present paper is closer in spirit to online decision-making with limited feedback than to average-accuracy-only benchmark chasing. The learner repeatedly decides whether a candidate update should be accepted, rejected, or frozen while supervisory information is sparse. We do not introduce a new partial-monitoring algorithm, but this viewpoint helps explain why binary correctness feedback can matter as a control primitive for an iterative adaptation heuristic [2].

3 Problem Setup

We start from a source-trained classifier $f_{\theta_0} : \mathcal{X} \to \Delta^{K-1}$ with a CIFAR-style ResNet-18 backbone [4]. At test time, the learner receives a stream partitioned into windows of size $W = 1024$. Our real benchmarks use both CIFAR-10 and CIFAR-100 label spaces. For stationary evaluation we use held-out CIFAR-10-C and CIFAR-100-C corruption panels [5]. For contamination evaluation, the same corruption streams are mixed with SVHN samples at contamination ratio

$$r \in \{0, 0.1, 0.2, 0.3, 0.5\}$$

using the deployed label space of the underlying CIFAR task [8].

The learner may adapt online, producing parameters θ_t from the current state and current window. We distinguish three information regimes:

1. **Frozen:** $\theta_t = \theta_0$ for all t.
2. **0-bit adaptation:** the learner updates from unlabeled target data only.
3. **Binary-feedback adaptation:** the learner may query a small subset and observe binary correctness supervision on those queried samples.

For an in-distribution queried sample i, let

$$b_i(\theta) = \mathbf{1}[\hat{y}_i(\theta) = y_i].$$

For a contamination sample outside the deployed label space, the queried correctness value is defined as 0 because the deployed classifier cannot make a correct in-label prediction on that sample. The nominal query-rate grid is

$$q \in \{0, 0.001, 0.005, 0.01, 0.02, 0.05, 0.10\}.$$

The key accounting point is that our main gate does not consume a single isolated bit detached from the update. It evaluates a proposed candidate update by comparing queried binary correctness before and after that update on the same queried subset. Operationally, each queried example therefore contributes a paired binary outcome to the gate statistic.

Our main metrics are in-distribution accuracy, gain over frozen, negative-transfer rate, worst-case gain over frozen across streams, expected calibration error (ECE), Brier score, overconfident error mass, accepted-update fraction, collapse rate, runtime, and peak memory. The analysis unit is a full stream run. A method incurs negative transfer on a stream if its in-distribution accuracy falls below the frozen baseline on that same stream.

4 Stylized Theory

The theory is intentionally stylized. Its role is not to model all neural-network dynamics, but to clarify why unlabeled evidence can be insufficient for safe update control and why paired binary feedback can help.

Proposition 1. *Consider two environments E_+ and E_- that induce the same unlabeled observations for a proposed update rule, but for which the same candidate update has positive gain over freezing in E_+ and negative gain over freezing in E_-. Then any deterministic accept-versus-freeze rule that depends only on unlabeled evidence must make the same decision in both environments, and therefore cannot be safe in both.*

Proof. If the unlabeled evidence is identical in the two environments, a deterministic 0-bit rule must output the same decision in both. Because the signed gain of the candidate update changes sign between E_+ and E_-, that shared decision is wrong in at least one environment.

This proposition does not say that label-free TTA is uniformly bad. It states a boundary on what unlabeled evidence alone can certify about the safety of a proposed update under contamination.

Let Q_t be the queried subset for a candidate update from θ_t to θ'_t and define

$$Z_i = b_i(\theta'_t) - b_i(\theta_t) \in \{-1, 0, 1\}, \qquad \widehat{\Delta}_t = \frac{1}{|Q_t|} \sum_{i \in Q_t} Z_i.$$

Our implemented gate accepts when $\widehat{\Delta}_t$ exceeds a threshold τ.

Proposition 2. *Assume the queried samples are conditionally independent for the purpose of estimating the paired gain $\Delta_t = \mathbb{E}[Z_i]$. If $\Delta_t \geq \tau + \gamma$ for some $\gamma > 0$, then*

$$\Pr(\widehat{\Delta}_t \leq \tau) \leq \exp\left(-\frac{|Q_t|\gamma^2}{2}\right).$$

Symmetrically, if $\Delta_t \leq \tau - \gamma$, then

$$\Pr(\widehat{\Delta}_t > \tau) \leq \exp\left(-\frac{|Q_t|\gamma^2}{2}\right).$$

Proof. Each Z_i is bounded in $[-1, 1]$. Hoeffding's inequality for bounded independent random variables gives

$$\Pr(\widehat{\Delta}_t - \Delta_t \leq -\gamma) \leq \exp\left(-\frac{|Q_t|\gamma^2}{2}\right)$$

and the corresponding upper-tail bound. Substituting $\Delta_t \geq \tau + \gamma$ or $\Delta_t \leq \tau - \gamma$ yields the result.

Corollary 1. *To make the false-accept and false-reject probabilities at most δ whenever the paired margin from threshold is at least γ, it is sufficient that*

$$|Q_t| \geq \frac{2\log(1/\delta)}{\gamma^2}.$$

This bound is only a stylized guide. It does not imply a target-accuracy guarantee for arbitrary neural models. It does, however, match the qualitative behavior of the implemented gate: once the effective query budget qW is large enough relative to the paired margin from threshold, the accept/reject decision becomes substantially more reliable.

5 Method

5.1 Candidate update

Our default 0-bit candidate is Tent [12]: entropy minimization on the current window with updates restricted to batch-normalization affine parameters and statistics [6]. Let θ_t denote the current safe state and X_t the current window. The candidate step produces θ'_t by applying one unlabeled Tent update on X_t.

5.2 Binary-feedback gate

The binary-feedback gate decides whether the candidate update is committed:

1. compute baseline predictions under θ_t;
2. choose queried indices Q_t with a fixed acquisition rule, defaulting to entropy;
3. snapshot the current model state;
4. apply one unlabeled Tent update on the whole window, producing candidate θ'_t;
5. evaluate queried binary correctness before and after the candidate on the same queried subset;
6. accept θ'_t only if the empirical paired gain exceeds threshold τ; otherwise restore the snapshot.

The gate statistic is

$$\widehat{\Delta}_t = \frac{1}{|Q_t|} \sum_{i \in Q_t} \big(b_i(\theta'_t) - b_i(\theta_t)\big).$$

The main experiments use threshold $\tau = 0.05$, one Tent step, and entropy-based query selection after development tuning. The implementation also supports rollback to the last accepted checkpoint.

Accounting note. The method therefore uses paired binary correctness supervision on queried examples, not a single free-standing bit detached from the update. This is exactly why we describe the method as binary-feedback-gated rather than retaining a stronger "one-bit" slogan.

5.3 Matched 0-bit controls

To test whether the gate helps only by making updates rarer, we compare against two matched-conservatism controls introduced in the shared protocol:

- **Self-Gate Tent (0-bit):** an unlabeled internal gate that uses only model-side signals to decide whether to commit the candidate update.
- **Throttle Tent (0-bit):** a stricter unlabeled update schedule tuned to reduce acceptance frequency without external feedback.

These controls are important because they separate two explanations: binary feedback might help because it simply makes the learner more conservative, or it might help because it provides genuinely external information that unlabeled heuristics do not have.

5.4 Compared methods

We compare four families of methods under a shared evaluation protocol:

1. **Frozen.**
2. **0-bit baselines:** Tent, a COME-style conservative entropy baseline, and the two matched-conservatism controls.
3. **Binary-feedback or low-label baselines:** the proposed binary gate, a BiTTA-style binary-feedback port, an ATTA-style active baseline, and an EATTA-style low-label baseline.
4. **Full-label reference:** a shared-protocol active baseline trained only on queried labels.

To ensure a shared protocol, we ported several baselines into a unified local evaluation harness with the same backbone family, source checkpoints, stream construction, query-rate grid, and reporting rules. Implementation and porting details are deferred to the online supplement.

6 Experimental Protocol

All methods are evaluated under one local protocol. We train three clean CIFAR-10 source models with seeds $\{0, 1, 2\}$ and clean test accuracies 0.9076, 0.9052, and 0.9041. We also train three clean CIFAR-100 source models with clean accuracies 0.7160, 0.7143, and 0.7167.

Development tuning uses designated development corruptions and then locks the default gate settings. The stationary evaluation uses ten held-out corruptions at severities 3 and 5. The contamination evaluation uses the same corruption set at severity 3 with SVHN contamination ratios

$$r \in \{0, 0.1, 0.2, 0.3, 0.5\}.$$

All real-data studies sweep

$$q \in \{0, 0.001, 0.005, 0.01, 0.02, 0.05, 0.10\}.$$

The results section emphasizes five pieces of evidence:

1. a synthetic ambiguity witness;
2. the stationary frontier, where no universal binary-feedback win is observed;
3. the contamination regime, where the safety story changes;
4. matched-conservatism controls;
5. robustness to noisy binary supervision and delayed commitment of already scored candidates.

The supervision-cost analysis later in the paper is derived from the same evaluation runs used throughout the study.

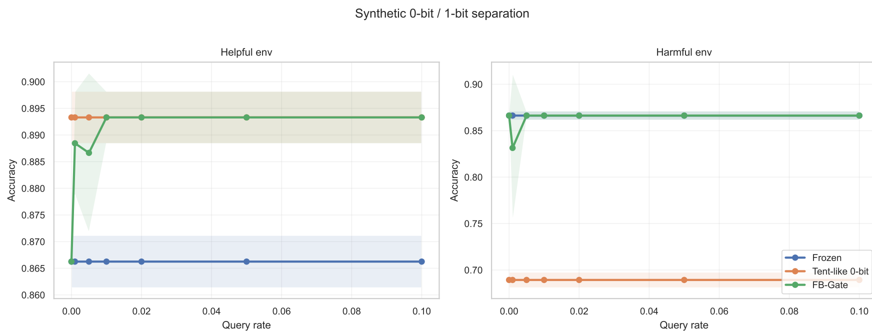

Fig. 1. Synthetic separation. The same 0-bit candidate update helps one environment and hurts the other. The paired binary gate accepts in the helpful environment and freezes in the harmful environment once the effective query budget is large enough.

7 Results

7.1 Synthetic separation

Figure 1 shows the stylized witness. In the helpful environment, the shared 0-bit candidate update gains $+0.0271$ accuracy over frozen. In the harmful environment, the same candidate loses -0.1770 and incurs negative transfer on every seed. The binary-feedback gate recovers the frozen-safe behavior in the harmful environment once the query rate reaches 0.005. At the smallest budget $q = 0.001$, the gate remains imperfect, which is exactly the finite-sample regime anticipated by the theory.

7.2 Stationary frontier

The stationary frontier provides the key negative result. On held-out CIFAR-10-C at $q = 0.01$, Tent reaches 0.7747 mean in-distribution accuracy, the COME-style 0-bit baseline reaches 0.7730, self-gated Tent reaches 0.7726, and the binary gate reaches 0.7703. On CIFAR-100-C the gap is larger: Tent reaches 0.5531 while the binary gate reaches 0.5304. No method exhibits observed negative transfer on this stationary panel.

This is an important scope boundary. The present results do not support a claim that binary feedback universally improves over strong 0-bit adaptation on average accuracy. In stationary corruption-only shift, the gate behaves mainly as a conservative wrapper that stays near the strongest 0-bit methods without surpassing them.

7.3 Contamination changes the safety story

The contamination benchmark is the main empirical test of the paper. Figure 3 shows the phase diagram in terms of negative-transfer rate. The main point is

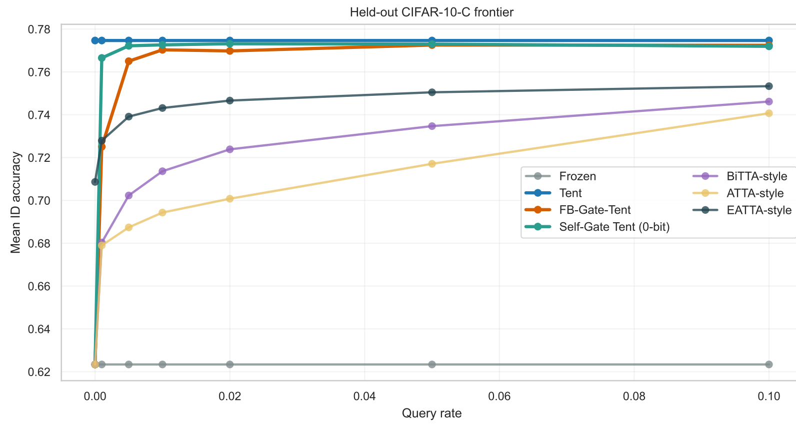

Fig. 2. Held-out CIFAR-10-C frontier on ten unseen corruptions at severities 3 and 5. Strong 0-bit methods remain best on mean accuracy. The binary-feedback gate stays close, but the stationary benchmark is not a universal binary-feedback win.

not that the binary gate has the highest mean accuracy. It does not. The point is that the contamination-time safety profile changes relative to strong 0-bit baselines.

At the primary operating point ($q = 0.01, r = 0.5$), Tent reaches 0.7884 accuracy on CIFAR-10-C plus SVHN with negative-transfer rate 0.1333, while the binary gate reaches 0.7728 with negative-transfer rate 0.0. On CIFAR-100-C plus SVHN, Tent reaches 0.5402 with negative-transfer rate 0.0667, while the binary gate reaches 0.5141 with negative-transfer rate 0.0. Thus the binary gate is not a highest-mean method, but it preserves positive gain over freezing while removing the observed negative-transfer events that remain for Tent at the hardest settings in our main grid.

7.4 Matched conservatism does not explain everything

A natural alternative explanation is that the benefit comes only from making Tent more conservative. Figure 4 addresses that possibility. Self-gated Tent nearly matches the binary gate on the stationary frontier, but under contamination it does not preserve the same safety profile. At ($q = 0.01, r = 0.5$), self-gated Tent has negative-transfer rate 0.10 on CIFAR-10-C plus SVHN and 0.0333 on CIFAR-100-C plus SVHN, whereas the binary gate remains at 0.0 on both. Throttle Tent likewise improves conservatism only partially and does not remove the high-contamination failures seen for fully label-free controls.

This is the strongest evidence in the current study that the external binary signal matters beyond internal confidence heuristics alone.

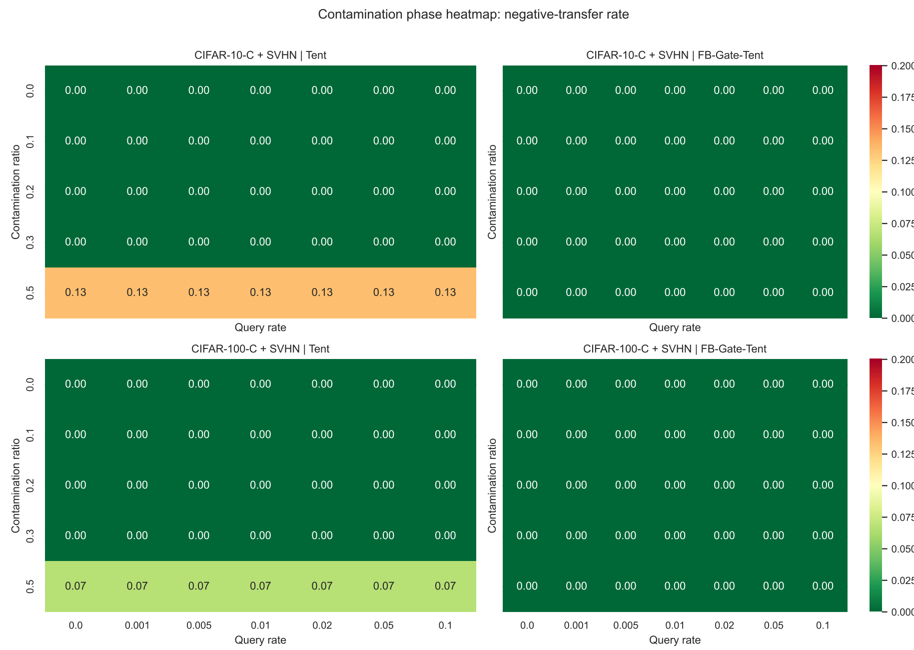

Fig. 3. Contamination phase heatmap. Each panel reports the observed negative-transfer rate across contamination ratio and query rate, rather than a formal theoretical phase boundary. The binary-feedback gate preserves a larger zero-negative-transfer region than Tent on both CIFAR-10-C plus SVHN and CIFAR-100-C plus SVHN.

7.5 Noise and delayed commitment

The robustness study shows that moderate binary-feedback noise is not catastrophic in this implementation. At $q = 0.01$ and $r = 0.5$, the binary gate retains positive gain over frozen for all tested noise levels on both datasets. On CIFAR-10-C plus SVHN, however, nonzero noise reintroduces a small negative-transfer rate of 0.0333; on CIFAR-100-C plus SVHN the observed negative-transfer rate remains 0.0 across the tested noise levels.

Delayed commitment is more harmful than moderate bit flips. At the same operating point, gain drops from 0.0293 to 0.0029 on CIFAR-10-C plus SVHN and from 0.0468 to 0.0088 on CIFAR-100-C plus SVHN when moving from immediate commitment to a five-window lag. The current implementation should be interpreted carefully: this lag study delays application of an already scored candidate update, rather than simulating labels arriving several windows later. We therefore present it as a robustness test for delayed commitment, not as a complete delayed-feedback model.

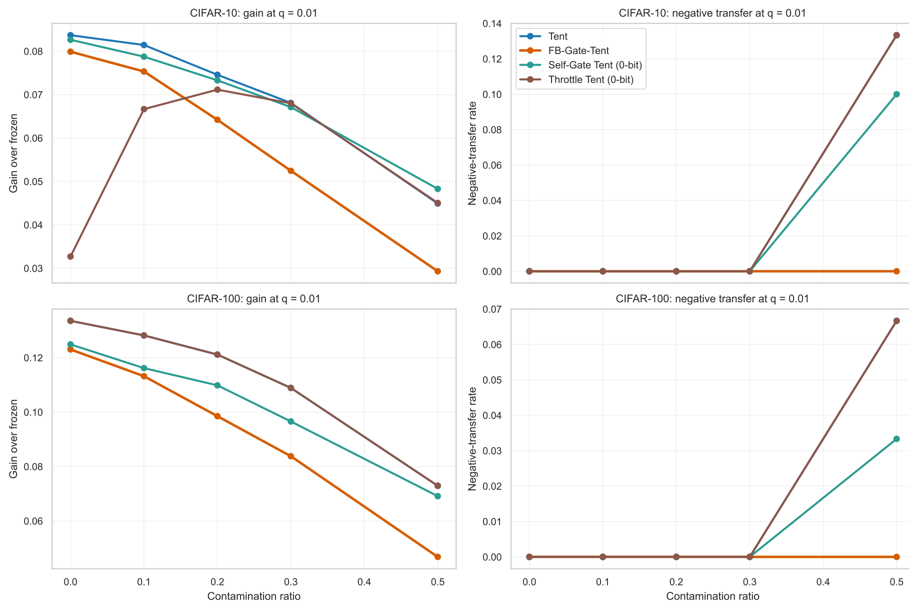

Fig. 4. Matched-conservatism comparison at $q = 0.01$. Self-gated and throttled 0-bit controls can mimic conservatism in stationary shift, but they do not recover the same contamination-time safety profile as the external binary veto.

7.6 A supervision-cost view

Table 1 reframes the primary contamination comparison by accounting for supervision events rather than pretending that all queried information is equivalent. Binary correctness events and label queries are reported as accounting units, not as equal-information units. The binary gate uses paired queried correctness evaluation, so at $q = 0.01$ it consumes 20 binary correctness events per 1,000 target samples. BiTTA-style adaptation consumes 10 binary correctness events per 1,000 target samples. EATTA-style low-label adaptation consumes 10 label queries per 1,000 target samples. Tent and the 0-bit controls consume no external supervision.

This table makes two points explicit. First, richer label-query methods can be stronger than the binary gate on mean accuracy, especially EATTA-style low-label adaptation. Second, the relevant scientific question here is not whether binary feedback dominates richer supervision, but whether a weaker supervision primitive can materially improve contamination-time safety relative to purely label-free controls. The answer from the current evidence is yes.

Table 1. Supervision-cost view at the primary contamination operating point ($q = 0.01, r = 0.5$). Binary correctness events and label queries are reported as accounting units, not as equal-information units. Event counts per 1k target samples are derived from the same evaluation runs used throughout the study.

Dataset	Method	Supervision	Events / 1k	Gain	NTR	Worst gain	Accept
CIFAR-10-C + SVHN	Frozen	0-bit	0.0	0.000	0.000	0.000	0.000
CIFAR-10-C + SVHN	Tent	0-bit	0.0	0.045	0.133	-0.013	1.000
CIFAR-10-C + SVHN	FB-Gate-Tent	binary correctness events	20.0	0.029	0.000	0.000	0.163
CIFAR-10-C + SVHN	Self-Gate Tent (0-bit)	0-bit	0.0	0.048	0.100	-0.005	0.443
CIFAR-10-C + SVHN	Throttle Tent (0-bit)	0-bit	0.0	0.045	0.133	-0.013	0.963
CIFAR-10-C + SVHN	COME-style 0-bit	0-bit	0.0	0.049	0.133	-0.009	1.000
CIFAR-10-C + SVHN	BiTTA-style	binary correctness events	10.0	0.007	0.400	-0.078	1.000
CIFAR-10-C + SVHN	EATTA-style	label queries	10.0	0.050	0.000	0.007	0.607
CIFAR-100-C + SVHN	Frozen	0-bit	0.0	0.000	0.000	0.000	0.000
CIFAR-100-C + SVHN	Tent	0-bit	0.0	0.073	0.067	-0.005	1.000
CIFAR-100-C + SVHN	FB-Gate-Tent	binary correctness events	20.0	0.047	0.000	0.000	0.170
CIFAR-100-C + SVHN	Self-Gate Tent (0-bit)	0-bit	0.0	0.069	0.033	-0.000	0.327
CIFAR-100-C + SVHN	Throttle Tent (0-bit)	0-bit	0.0	0.073	0.067	-0.005	1.000
CIFAR-100-C + SVHN	COME-style 0-bit	0-bit	0.0	0.077	0.033	-0.001	1.000
CIFAR-100-C + SVHN	BiTTA-style	binary correctness events	10.0	0.026	0.167	-0.053	1.000
CIFAR-100-C + SVHN	EATTA-style	label queries	10.0	0.094	0.000	0.025	1.000

8 Discussion and Limitations

The present study supports a narrow claim. It does not show that binary correctness feedback universally improves mean accuracy over strong label-free test-time adaptation. On held-out stationary CIFAR-10-C and CIFAR-100-C, Tent and strong 0-bit controls remain the best mean-accuracy baselines. The contamination setting changes the story: under CIFAR-C plus SVHN contamination, the binary-feedback gate preserves positive gain over freezing while removing the observed negative-transfer events that remain for Tent and matched-conservatism 0-bit controls at the hardest settings in the main grid.

This narrower conclusion also clarifies the role of richer supervision. Low-label active baselines, especially the EATTA-style port, can outperform the binary gate on mean accuracy while also retaining a safe profile. When label queries are affordable, richer low-label methods can be preferable if the goal is to maximize mean accuracy. Our question is narrower: whether weaker binary correctness supervision already changes contamination-time update control relative to purely label-free adaptation. That does not contradict the main claim. The point of the paper is that even a weaker supervision primitive can materially improve update safety under contamination.

Several limitations should be stated explicitly. First, some non-core baselines are shared-protocol ports rather than end-to-end official-runner reproductions. Second, the contamination source is SVHN only, so we do not claim that the exact phase boundaries transfer unchanged to every open-world contaminant. Third, the current lag study is a delayed-commitment setting rather than a full delayed-label-arrival simulation. Finally, the theory is stylized and explains the information boundary and the gate's finite-sample behavior, rather than giving worst-case guarantees for arbitrary neural dynamics.

9 Conclusion

Streaming test-time adaptation can be understood as an iterative online heuristic whose main difficulty under contamination is update control under limited information. From that perspective, the contribution of binary correctness feedback is not a universal average-accuracy advantage. It is a small external veto primitive for deciding whether a proposed update should be accepted, rejected, or frozen.

The current evidence supports that narrow claim. Purely unlabeled evidence can be ambiguous about update safety, stationary CIFAR-C does not show a universal binary-feedback win, and contamination changes the safety story in favor of the binary-feedback gate. This makes binary correctness feedback a useful control signal for conservative deployment under contamination, even when stronger low-label methods remain preferable for maximizing mean accuracy.

References

1. Bendale, A., Boult, T.E.: Towards open set deep networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1563–1572 (2016), https://openaccess.thecvf.com/content_cvpr_2016/html/Bendale_Towards_Open_Set_CVPR_2016_paper.html
2. Cesa-Bianchi, N., Lugosi, G.: Prediction, Learning, and Games. Cambridge University Press (2006)
3. Gui, S., Li, X., Ji, S.: Active test-time adaptation: Theoretical analyses and an algorithm. In: The Twelfth International Conference on Learning Representations (2024), https://openreview.net/forum?id=YHUGlwTzFB
4. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2016)
5. Hendrycks, D., Dietterich, T.: Benchmarking neural network robustness to common corruptions and perturbations. In: International Conference on Learning Representations (2019)
6. Ioffe, S., Szegedy, C.: Batch normalization: Accelerating deep network training by reducing internal covariate shift. In: International Conference on Machine Learning (2015)
7. Lee, T., Chottananurak, S., Kim, J., Shin, J., Gong, T., Lee, S.J.: Test-time adaptation with binary feedback. In: International Conference on Machine Learning (2025)
8. Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A.Y.: Reading digits in natural images with unsupervised feature learning. NeurIPS Workshop on Deep Learning and Unsupervised Feature Learning (2011)
9. Niu, S., Wu, J., Zhang, Y., Chen, Y., Zheng, S., Zhao, P., Tan, M.: Efficient test-time model adaptation without forgetting. In: Proceedings of the 39th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 162, pp. 16888–16905. PMLR (2022), https://proceedings.mlr.press/v162/niu22a.html
10. Niu, S., Wu, J., Zhang, Y., Wen, Z., Chen, Y., Zhao, P., Tan, M.: Towards stable test-time adaptation in dynamic wild world. In: International Conference on Learning Representations (2023)

11. Sun, Y., Wang, X., Liu, Z., Miller, J., Efros, A., Hardt, M.: Test-time training with self-supervision for generalization under distribution shifts. In: Proceedings of the 37th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 119, pp. 9229–9248. PMLR (2020), https://proceedings.mlr.press/v119/sun20b.html
12. Wang, D., Shelhamer, E., Liu, S., Olshausen, B., Darrell, T.: Tent: Fully test-time adaptation by entropy minimization. In: International Conference on Learning Representations (2021)
13. Wang, G., Ding, C.: Effortless active labeling for long-term test-time adaptation. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 25633–25642 (2025)
14. Wang, Q., Fink, O., Van Gool, L., Dai, D.: Continual test-time domain adaptation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 7201–7211 (2022), https://arxiv.org/abs/2203.13591
15. Zhang, M., Levine, S., Finn, C.: Memo: Test time robustness via adaptation and augmentation. In: NeurIPS Workshop on Distribution Shifts (2021), https://openreview.net/forum?id=vn74m_tWu8O
16. Zhang, Q., Bian, Y., Kong, X., Zhao, P., Zhang, C.: Come: Test-time adaption by conservatively minimizing entropy. In: International Conference on Learning Representations (2025)